Assignment: 9
Points: 30
Due Date: 05/08/2013 12:00 PM

(3 points each)
Chapter 9:
pp. 236-237 (4th edition), 238-239 (5th edition), Nos. 2*, 3, 4, 5, 6

Design Turing machines for the following languages by drawing the transition graph for each Turing machine. Assume the input alphabet is {a, b}.

1. A TM that accepts the languages of all strings that contain the substring *bbb*.
2. A TM that accepts the languages of all strings that do not contain the substring *bbb*.
3. A TM that accepts the language of odd length palindromes.
4. A TM that accepts all strings with more *a*'s than *b*'s.
5. A TM that accepts the language $\{a^n b^{n+1} : n \geq 0\}$.
6. A TM that accepts the language $\{a^n b^{2n} : n \geq 0\}$.

You should hand-in only the problems without '*'. Hand written solutions will **NOT** be graded.